



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Internet of Things (IoT) Fundamental for Beginners

TGS Ref No: TGS-2020504020

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 14

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
12	1 March 2025	Previous release — ThingSpeak/ESP8266/Node-RED based courseware.	Dr. Alfred Ang
13	6 July 2026	Revamped to the IoTFlow platform (iot.tertiaryinfotech.com): 6 labs covering device registration, MQTT/HTTP telemetry, virtual-pin remote control, dashboards and n8n + AI automation.	Dr. Alfred Ang
14	6 July 2026	Labs updated to the official IoTFlow Python client (pip install iotflow): HTTP send/virtual_write, the telemetry_upload.py MQTT example, and @on_command control handlers.	Dr. Alfred Ang

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Course Information

Course Title	Internet of Things (IoT) Fundamental for Beginners
WSQ Course Reference	TGS-2020504020
Skills Framework TSC	Internet of Things Application (PTP-TEM-3002-1.1)
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on the IoTFlow IoT platform (https://iot.tertiaryinfotech.com) using the official Python client (pip install iotflow) and n8n
Trainer	Dr. Alfred Ang
Course Page	https://www.tertiarycourses.com.sg/wsq-internet-of-things-iot-fundamental-for-beginners.html

Skills Framework — TSC Abilities and Knowledge

This course is mapped to the Skills Framework TSC Internet of Things Application (PTP-TEM-3002-1.1):

- A1: Conduct briefings on the uses and functions of IoT technologies adopted by the organisation
- A2: Review IoT testing results and identify areas for improvement
- A3: Integrate information from multiple data sources
- A4: Review data to produce insights of business value
- K1: Concept of Internet of Things (IoT)
- K2: Types and functionalities of IoT devices
- K3: Types of circuits and sensors within devices
- K4: Types of wireless communication technologies
- K5: Data analytics techniques
- K6: Concept of cybersecurity

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Understand the uses and functions of IoT technologies
- LO2: Post sensor data to cloud for IoT review
- LO3: Control devices from cloud data sources
- LO4: Data analytics and visualization on cloud to gain business insight

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on IoT platform tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — IoT Overview & Collecting Data to the Cloud

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, learning outcomes, course outline, ground rules and mandatory digital attendance (AM)
10:00–11:00	60 min	Topic 1 — Overview of Internet of Things (IoT): what is IoT, IoT devices, sensors and actuators, triggers and events (concepts + demo)
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Topic 1 (continued) — wireless communication technologies, IoT applications and use cases. Introduction to our IoT platform IoTFlow (iot.tertiaryinfotech.com): how it works — connect, visualise & control, automate with n8n
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 2 — Collect and Post Data to Cloud: cloud computing, IoT platforms, MQTT and REST API, the IoTFlow Python client (pip install iotflow) (concepts). Hands-on: Lab 1: Register Your First Device on IoTFlow
15:30–15:45	15 min	Tea break
15:45–17:45	120 min	Hands-on: Lab 2: Send Sensor Readings to the Cloud (HTTP & MQTT)
17:45–18:30	45 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Remote Control, Analytics, n8n Automation & Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory

9:45–11:00	75 min	digital attendance (AM) Topic 3 — Read Data and Remote Control from Cloud: REST API, MQTT subscribe, virtual pins (concepts). Hands-on: Lab 3: Read Device Data from the Cloud (REST API & MQTT)
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 4: Remote Control a Device with Dashboard Virtual Pins. Alerts and trigger rules
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	Topic 4 — IoT Data Analytics and Visualization: dashboards, analytics, cybersecurity (concepts). Hands-on: Lab 5: Build a Real-Time IoT Dashboard
15:00–15:15	15 min	Tea break
15:15–16:15	60 min	Hands-on: Lab 6: Automate with n8n — Triggers, Workflows and AI
16:15–16:30	15 min	Course feedback and TRAQOM survey; Briefing for Assessment; Assessment digital attendance
16:30–17:30	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book
17:30–18:30	60 min	Practical Performance (PP) — hands-on IoT platform tasks, 1 hour, open book. End of class

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to topics)

Topic	Abilities / Knowledge	Labs
Topic 01: Overview of Internet of Things (IoT)	A1, K1, K4	Concepts and demonstrations (no lab)
Topic 02: Collect and Post Data to Cloud	A2, K3	Lab 1: Register Your First Device on IoTFlow; Lab 2: Send Sensor Readings to the Cloud (HTTP & MQTT)
Topic 03: Read Data and Remote Control from Cloud	A3, K2	Lab 3: Read Device Data from the Cloud (REST API & MQTT); Lab 4: Remote Control a Device with Dashboard Virtual Pins
Topic 04: IoT Data Analytics and Visualization	A4, K5, K6	Lab 5: Build a Real-Time IoT Dashboard; Lab 6: Automate with n8n — Triggers, Workflows and AI